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Department of Information Technology

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Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 8: Security as a Service (SECaaS)

Aim: To study and Implement Security as a Service on AWS using ADDS service. (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to –
- to understand Cloud security at AWS
- AWS Directory Service for identity management
- deploy AWS administration and security services.

Lab objective mapped: ITDO8024.4: To learn cloud deployment model and experiment security services over different cloud platforms.

Prerequisite:
An AWS Account

Requirements: Cloud Login**,** Desktop, Browser, Internet etc.

Pre-Experiment Theory:

AWS Directory Service
AWS Directory Service is a managed service that provides directories that contain information about your organization, including users, groups, computers, and other resources.

Windows Server Manager
Microsoft Windows Server Manager is a tool to view and manage server roles and make configuration changes. Server Manager allows administrators to manage local and remote servers without requiring physical access to the servers or enabling Remote Desktop Protocol connections.

Procedure

AWS Managed Microsoft AD prerequisites

- To create a AWS Managed Microsoft AD directory, you need a VPC with the following:
- At least two subnets. Each of the subnets must be in a different Availability Zone.
- The VPC must have default hardware tenancy.
- You cannot create a AWS Managed Microsoft AD in a VPC using addresses in the 198.18.0.0/15 address space.

To create an AWS Managed Microsoft AD directory

1. AWS Directory Service console navigation pane, choose Set up directory.
2. Select directory type : choose AWS Managed Microsoft AD, and then click Next.
3. Enter directory information page :
4. Edition :Choose Standard Edition.
5. Directory DNS name :The fully qualified name (FQDN) for the directory, *beit_exp5.com*.
6. Directory NetBIOS name : optional.
7. Directory description : An optional description for the directory.
8. Admin password : The password for the directory administrator. The directory creation process creates an administrator account with the user name **Admin**. (The password cannot include the word "admin."). Read rules for setting up the directory administrator password.
9. Confirm password
10. Choose VPC and subnets page (refer to VPC creation procedure) :
11. The VPC for the directory.
12. Subnets: Choose the subnets for the domain controllers. The two subnets must be in different Availability Zones.
13. choose Next.
14. On the Review & create page : review the directory information. Choose Create directory.
15. *Creating the directory takes 20 to 40 minutes (What gets created).*
16. *Once created, the Status value changes to Active (green in color).*

To seamlessly join a Windows EC2 instance

1. Go to Amazon EC2 console
2. From the Region selector in the navigation bar, choose the same Region as the existing directory.
3. Launch Instance.
4. AMI : win server 2019 free tier full edition
5. instance type : t2.micro
6. Next: Configure Instance Details.
7. Step 3 page:

8. Network : VPC that your directory was created in.
9. Subnet : choose one of the public subnets in your VPC. (The subnet that you choose must have all external traffic routed to an internet gateway.)
10. Domain join directory : choose your domain from the list.
11. IAM role:
    a. Choose Create role.
    b. Under Select type of trusted entity, choose AWS service.
    c. service role will use : choose EC2.
    d. Select your use case : choose EC2
    e. Next: Permissions.
    f. In the list of policies : 1. AmazonSSMManagedInstanceCore 2.AmazonSSMDirectoryServiceAccess policies
    g. Choose Next: Tags(Optional)
    h. choose Next: Review.
    i. Role name : EC2DomainJoin
    j. Choose Create role.

12. Go back to the Step 3 page. For IAM role, choose the refresh icon next to IAM role.Choose created role and leave the rest of the settings with default values
13. choose Next: Add Storage.
14. On both the Step 4 and Step 5 pages, leave the default settings or make changes as needed, and then choose the Next buttons.
15. On the Step 6 page, select a security group for the instance that has been configured to allow remote access to the instance from your network, and then choose Review and Launch.
16. Step 7 page, choose Launch, select a key pair, and then choose Launch Instance..

To install the Active Directory administration tools on Windows Server 2012 through Windows Server 2019

1. Open Server Manager from the Start screen by choosing Server Manager.
2. In the Server Manager Dashboard, choose Add roles and features,In the Add Roles and Features Wizard choose Installation Type, select Role-based or feature-based installation, and choose Next.
3. Under Server Selection, make sure the local server is selected, and choose Features in the left navigation pane.
4. In the Features tree, open Remote Server Administration Tools, Role Administration Tools, select AD DS and AD LDS Tools, scroll down and select DNS Server Tools, and then choose Next.
5. Review the information and choose Install.
6. When the feature installation is finished, the Active Directory Domain Services and Active Directory Lightweight Directory Services Tools are available on the Start screen in the Administrative Tools folder.

To create a user

1. Open the Active Directory Users and Computers tool. There is a shortcut to this tool in the Administrative Tools folder.
2. On the Action menu, click New, and then click User to open the new user wizard.
3. On the first page of the wizard, enter the values for the following fields, and then click Next.
4. First name : student name
5. Last name : student last name
6. User logon name : first name
7. type a temporary password in Password and Confirm Password. Make sure the User must change password at next logon option is selected. None of the other options should be selected. Click Next.
8. On the third page of the wizard, verify that the new user information is correct and click Finish.
9. The new user will appear in the Users folder.

Create a new VPC

1. VPC console
2. choose Launch VPC Wizard.
3. Under Step 1: Select a VPC Configuration, on VPC with a Single Public Subnet, choose Select.
4. Enter the following information into the wizard and choose Create VPC.

IP CIDR block
10.100.0.0/16

VPC name
BEIT_EXP5_VPC

Public subnet
10.100.0.0/24

Availability Zone
us-east-1a

Subnet name
exp5_Subnet1

Enable DNS hostnames
Leave default selection

Hardware tenancy

Default

It takes several minutes for the VPC to be created.

Post-Experiments Exercise

1. Name three directory types that AWS supports? (to be written in hand)
2. List any five permissions an Admin account has to perform the administrative activities for your organizational unit (OU)? (to be written in hand)

Extended Theory:
What is Windows Server Manager?(soft copy)

.**Results/Calculations/Observations:**

Post Experimental Exercise-
How to reset password for already created user in AWS AD service?

Conclusion:
1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:
[1] [Online]https://learn.microsoft.com/en-us/azure/app-service
[2] [Online]https://learn.microsoft.com/en-us/azure/app-service/quickstart-nodejs

Search [Alt+S] United States (N. Virginia) Account ID: 9895-7055-1455 DURVAA

VPC > Your VPCs > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

- (•) VPC only
- ( ) VPC and more

Name tag - *optional*
Creates a tag with a key of 'Name' and a value that you specify.

Durva_vpc1

IPv4 CIDR block Info

- (•) IPv4 CIDR manual input
- ( ) IPAM-allocated IPv4 CIDR block

IPv4 CIDR

12.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

- (•) No IPv6 CIDR block
- ( ) IPAM-allocated IPv6 CIDR block
- ( ) Amazon-provided IPv6 CIDR block
- ( ) IPv6 CIDR owned by me

VPC > Your VPCs > Create VPC

IPv6 CIDR block Info

- (•) No IPv6 CIDR block
- ( ) IPAM-allocated IPv6 CIDR block
- ( ) Amazon-provided IPv6 CIDR block
- ( ) IPv6 CIDR owned by me

Tenancy Info

Default

VPC encryption control ($) Info
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. Additional charges apply

- (•) None
- ( ) Monitor mode — See which resources in your VPC are unencrypted but allow the creation of unencrypted resources.
- ( ) Enforce mode — Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - *optional*	
Name	Durva_vpc1	Remove tag

Add tag

You can add 49 more tags

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs: School

Search [Alt+S] United States (N. Virginia) DURVAA (9895-7055-1455) DURVAA

VPC > Your VPCs

VPC dashboard

AWS Global View

Filter by VPC

- Virtual private cloud
  - Your VPCs
  - Subnets
  - Route tables
  - Internet gateways
  - Egress-only internet gateways
  - Carrier gateways
  - DHCP option sets
  - Elastic IPs
  - Managed prefix lists
  - NAT gateways
  - Peering connections
  - Route servers
- Security
  - Network ACLs
  - Security groups

You successfully deleted vpc-0b7f8133e806ecab8 / Durva_vpc1

Your VPCs

VPCs | VPC encryption controls

Your VPCs (2) Info

Last updated less than a minute ago | Actions | Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Publi
Durva-vpc1	vpc-0f83e49d3c3614f2c	Available	–	–	Off
–	vpc-08b493ec7bb106ba9	Available	–	–	Off

vpc-0b7f8133e806ecab8 / Durva_vpc1

Details | Resource map | CIDRs | Flow logs | Tags | Integrations

Details

VPC ID vpc-0b7f8133e806ecab8	**State** Available	**Block Public Access** Off	**DNS hostnames** Disabled
DNS resolution Enabled	**Tenancy** default	**DHCP option set** dopt-0cd69d836abadee93	**Main route table** rtb-089f1507eb723a2c1
Main network ACL	**Default VPC**	**IPv4 CIDR**	**IPv6 pool**

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateSubnet:

VPC > Subnets > Create subnet

Create subnet Info

VPC

VPC ID
Create subnets in this VPC.
vpc-0f83e49d3c3614f2c (Durva-vpc1)

Associated VPC CIDRs

IPv4 CIDRs
12.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
public-subnet
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateSubnet:

VPC > Subnets > Create subnet

The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1a)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
12.0.0.0/16

IPv4 subnet CIDR block
12.0.1.0/24 256 IPs

▼ **Tags - *optional***

Key	Value - *optional*	
Name	public-subnet	Remove

Add new tag
You can add 49 more tags.
Remove

Add new subnet

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#AttachInternetGateway:internetGatewayId=igw-0cd77a06a4659520f

VPC > Internet gateways > Attach to VPC (igw-0cd77a06a4659520f)

The following internet gateway was created: igw-0cd77a06a4659520f - Durva-gateway. You can now attach to a VPC to enable the VPC to communicate with the internet. Attach to a VPC

Attach to VPC (igw-0cd77a06a4659520f) Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.
vpc-0f83e49d3c3614f2c

▼ **AWS Command Line Interface command**

You can perform the same actions on this page by using the AWS Command Line Interface (CLI) tools. Learn more.

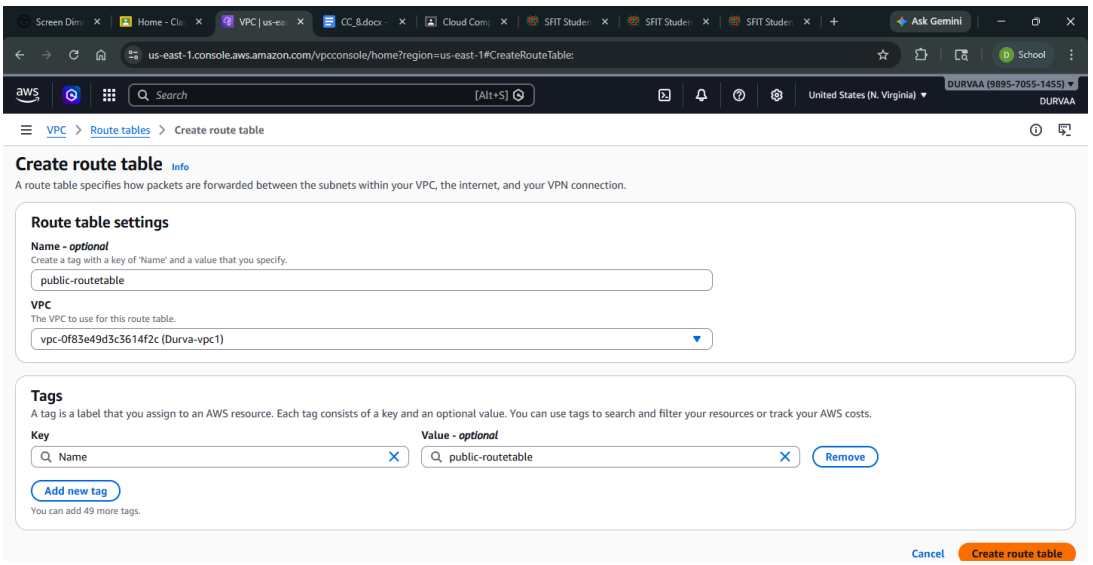
Platform
Choose the platform from which you'll be running this command. The command parameters may be specified differently depending on the platform. Learn more about specifying parameter values.
Linux/Unix/OS X

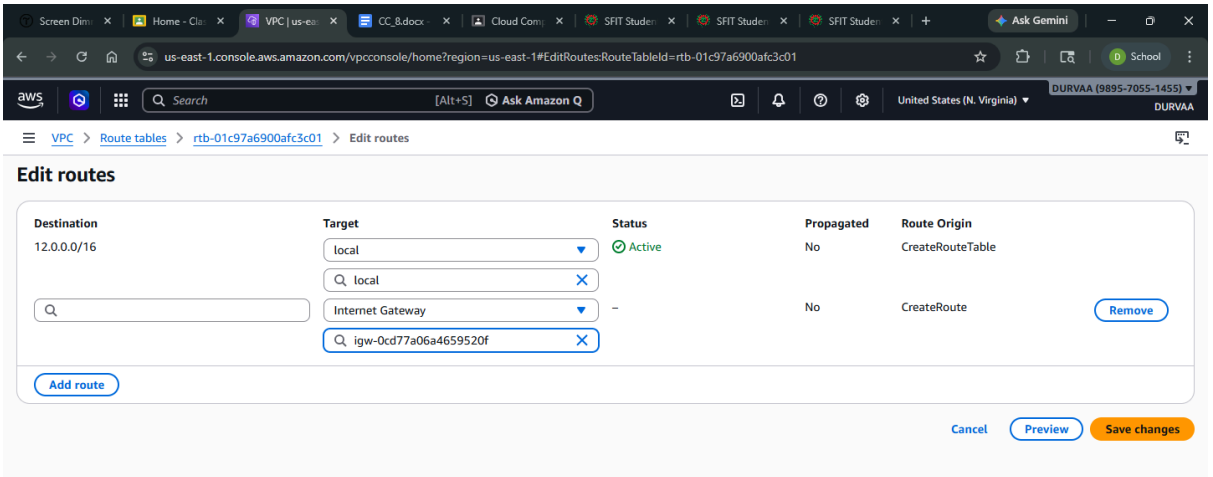
CLI command
If you're using the AWS CLI tools, you can copy and paste this command - which includes the parameters you specified on this page - into your command line prompt or terminal. Learn more about the available AWS CLI commands.

```
aws ec2 attach-internet-gateway --vpc-id "vpc-0f83e49d3c3614f2c" --internet-gateway-id "igw-0cd77a06a4659520f" --region us-east-1
```

Copy

Cancel Attach internet gateway





Extended Theory:

1. **What is Windows Server Manager?(soft copy)**

ANS:

1. **Windows Server Manager** is a management tool in Microsoft Windows Server used to manage and monitor servers.
2. It provides a **centralized dashboard** to control multiple servers from one place.
3. Helps administrators **install, configure, and manage server roles and features**.
4. Allows adding roles like **Web Server (IIS), DNS, Active Directory, etc.**
5. Supports **remote server management**, so multiple servers can be handled from a single system.
6. Displays **server status, performance, and alerts** in real time.
7. Helps in **troubleshooting and monitoring system health**.
8. Provides **easy access to tools like PowerShell, Event Viewer, and Services**.